■ BambooSense

Automated Bamboo Growth Monitoring · Biomass & Carbon Assessment Report

Report Summary

Generated	08 May 2026	**Model**	YOLOv8s — bamboo fine-tuned
Images Processed	7	**Total Culms**	29
Health Breakdown	Yellow: 20 · Green: 9	**Avg Height**	137.72 cm
Total Biomass	2354.183 kg	**Avg Diameter**	4.04 cm

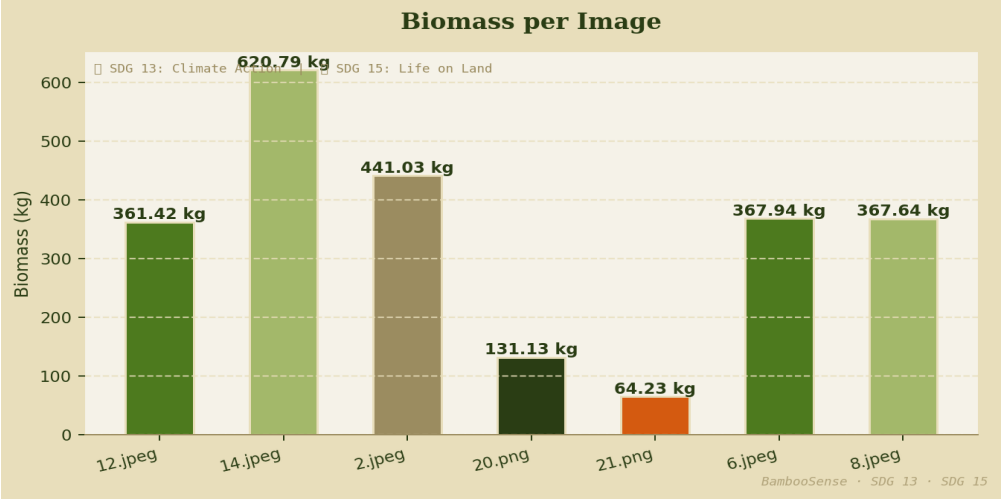
■ Total CO■ Sequestered: 392.868 kg (0.39287 tonnes) · ≈ 18.0 trees annual absorption equivalent

Carbon factor: 0.25 kg CO■ / kg dry biomass (INBAR TR-37 / IPCC Tier 1) · Dry culm carbon penalty applied (×0.20)

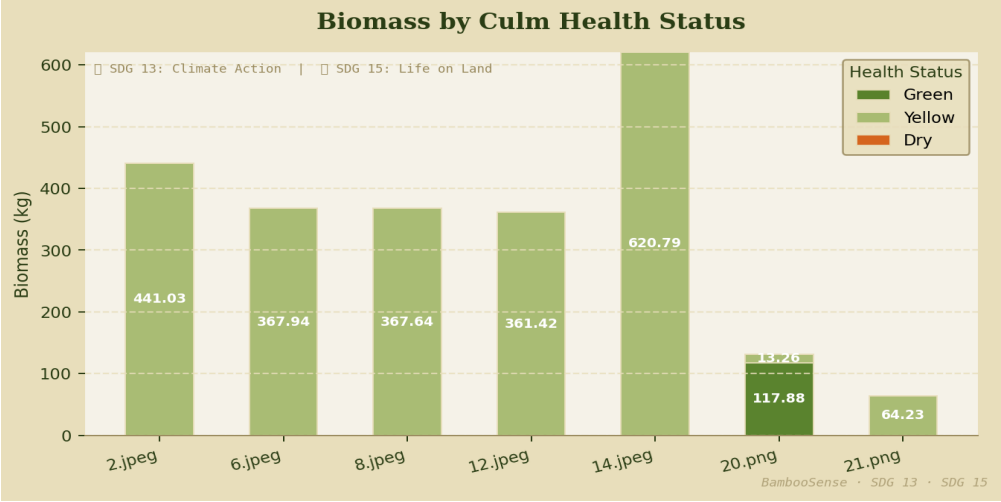
SDG Alignment

SDG 13 (Climate Action): Bamboo forests are a measurable, low-cost natural carbon sink. SDG 15 (Life on Land): BambooSense supports sustainable forest management and biodiversity monitoring. SDG 11 (Sustainable Cities): Bamboo biomass data informs construction material supply chains.

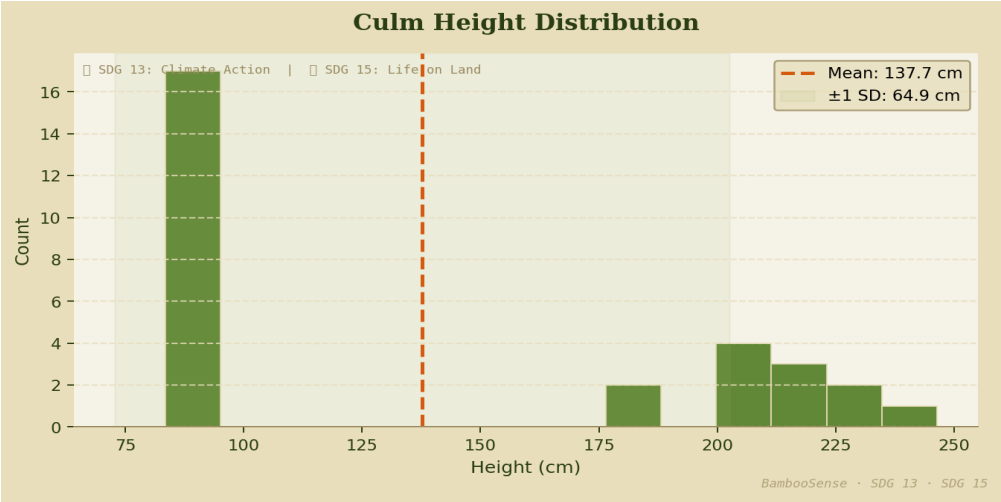
Biomass per Image



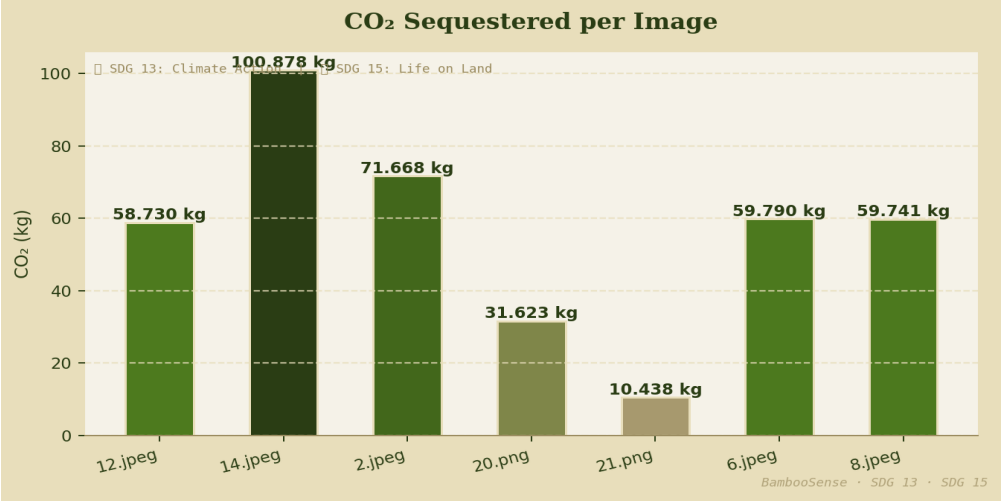
Biomass by Health Status



Culm Height Distribution



CO■ Sequestered



Detailed Culm Measurements

Image	Culm #	Health	Height (cm)	Diameter (cm)	Biomass (kg)	CO■ (kg)
2.jpeg	1	Yellow	221.44	5.99	154.904	25.172

Image	Culm #	Health	Height (cm)	Diameter (cm)	Biomass (kg)	CO■ (kg)
2.jpeg	2	Yellow	210.07	4.15	70.431	11.445
2.jpeg	3	Yellow	210.07	7.26	215.696	35.051
6.jpeg	1	Yellow	217.45	6.97	205.906	33.46
6.jpeg	2	Yellow	210.87	6.28	162.032	26.33
8.jpeg	1	Yellow	228.22	7.66	261.097	42.428
8.jpeg	2	Yellow	178.75	5.53	106.542	17.313
12.jpeg	1	Yellow	228.02	5.93	156.456	25.424
12.jpeg	2	Yellow	216.45	6.97	204.962	33.306
14.jpeg	1	Yellow	201.09	7.14	199.975	32.496
14.jpeg	2	Yellow	246.38	7.66	281.866	45.803
14.jpeg	3	Yellow	177.55	6.33	138.947	22.579
20.png	1	Green	85.98	2.53	16.563	4.141
20.png	2	Green	85.58	2.19	12.297	3.074
20.png	3	Green	85.38	2.36	14.281	3.57
20.png	4	Green	86.18	1.79	8.241	2.06
20.png	5	Green	85.78	2.42	15.057	3.764
20.png	6	Yellow	85.38	2.82	13.259	2.155
20.png	7	Green	85.58	2.13	11.658	2.914
20.png	8	Green	85.78	1.55	6.222	1.556
20.png	9	Green	85.38	2.19	12.268	3.067
20.png	10	Green	85.58	2.88	21.289	5.322
21.png	1	Yellow	83.79	2.13	7.419	1.206
21.png	2	Yellow	83.39	2.36	9.066	1.473
21.png	3	Yellow	84.79	2.59	11.104	1.804
21.png	4	Yellow	84.79	2.07	7.107	1.155
21.png	5	Yellow	84.59	2.13	7.489	1.217
21.png	6	Yellow	84.79	2.25	8.34	1.355
21.png	7	Yellow	84.79	2.88	13.709	2.228